Requirements-based testing of deep neural networks.

RBT4DNN · A critical review of its empirical evidence and two focused additional analyses.

SEMINAR	COURSE LEADERSHIP	PRESENTATION
AI Testing Landscape: Ensuring Robustness through Advanced Testing Methods	Prof. Dr. Alexander Pretschner Supervisors: Vivek V. Vekariya · Simon Speth	Caspar Breloh 17 July 2026

Three questions guide the argument.

01 · NEED

Why do neural-network tests need requirements?

PROBLEM + BACKGROUND

02 · METHOD

How does RBT4DNN generate and evaluate targeted tests?

APPROACH + METHODOLOGY + RESULTS

03 · EVIDENCE

How convincing—and scalable—is the resulting evidence?

CRITIQUE + ADDITIONAL ANALYSES

These questions follow the seminar brief while keeping one continuous evidence argument.

High average accuracy does not answer a specific requirement.

CONVENTIONAL TESTING

Known behavior

A specification maps an input condition to an expected output. Test oracles can check the result directly.

DNN TESTING

Learned behavior

The model is inferred from examples. Global accuracy summarizes a distribution, not every meaningful scenario.

THE MISSING LINK

Semantic scenarios

Developers need evidence for statements such as "a very thick 7 must still be classified as 7."

GLOBAL METRIC
How often is the complete test set correct?

→

SEMANTIC SLICE
Which rare inputs does the score contain?

→

REQUIREMENT VERDICT
Does the model satisfy this specific contract?

[1] §1–§2.5 · "model under test" means the evaluated neural network

Existing generators rarely start from the requirement itself.

Neuron coverage, image transformations, and search-based generators can produce many tests—without proving that those tests represent the scenario a stakeholder cares about.

INPUT	Requirement	Structured natural language defines a semantic precondition and an expected outcome.
METHOD	RBT4DNN	Fine-tune a generator on images that satisfy that precondition.
OUTPUT	Targeted tests	Run the learned component and evaluate the requirement-specific oracle.

[1] §1, §2.5 and §3 · authors claim the first requirement-driven DNN test-input generator

Four domains test whether one workflow survives increasingly ambiguous semantics.

MNIST · 7 REQUIREMENTS

MNIST artifact samples
Source-linked in interactive deck

MNIST

Measured height, thickness, and lean. A controlled starting point.

Output: digit class

CELEBA-HQ · 7 REQUIREMENTS

CelebA-HQ artifact samples
Source-linked in interactive deck

CelebA-HQ

Ambiguous attributes labeled by visual question answering or humans.

Output: eyeglasses decision

SGSM · 7 REQUIREMENTS

SGSM artifact samples
Source-linked in interactive deck

SGSM

Synthetic driving scenes with spatial relations linked to steering and acceleration.

Output: numeric control

IMAGENET · 4 REQUIREMENTS

ImageNet artifact samples
Source-linked in interactive deck

ImageNet

Animal morphology linked to allowed classes in the WordNet taxonomy.

Output: class set

[1] Table 2 and §4.1.1–§4.1.3 · samples linked from the pinned public artifact

MNIST M3 artifact samples
Source-linked in interactive deck

A requirement defines both relevance and correctness.

If the digit is a 7 and is very thick, **then** the model shall label it as 7.

PRECONDITION	POSTCONDITION
Which images count as relevant tests?	Which model outputs count as acceptable?

[1] Table 2, M3 · a finite passing suite cannot prove the universal requirement

RBT4DNN turns one requirement into a targeted test campaign.

01 · LABEL

Pixels → terms

Attach domain facts such as very thick, eyeglasses, or same lane.

→

02 · FILTER

Terms → subset

Keep original images satisfying one requirement precondition.

→

03 · ADAPT

Subset → LoRA

Fine-tune FLUX with a small adapter for that precondition.

→

04 · TEST

Images → evidence

Generate tests, run the model under test, and check the required output.

M3 LABELS	FILTERED SUBSET	TARGETED GENERATOR	EVIDENCE
digit = 7 · thickness = very thick	only matching MNIST images	one M3 LoRA adapter	precondition match + output pass or fail

[1] Fig. 2 and §3 · symbols are translated into an explicit input/output contract

Glossary terms make visual requirements operational.

GLOSSARY TERM

A **named visual property** that can be assigned to an image: “very thick,” “same lane,” or “eyeglasses.”

01 · DEFINE

Authors specify the glossary

Requirement concepts are manually turned into a finite vocabulary of visual properties.

TERM CREATION IS NOT AUTOMATIC

02 · LABEL ORIGINAL DATA

Labels come from domain-specific signals

MNIST measurements · SGSM scene graphs · MiniCPM visual Q&A for CelebA-HQ and ImageNet.

CELEBA ALSO HAS HUMAN LABELS

03 · USE THE LABELS

Filter first; check later

Labels select the LoRA training subset. Separate binary classifiers then check glossary terms on generated images.

PRECONDITION MATCH · RQ1 / RQ3

Boundary: separate training labels and evaluation classifiers reduce direct reuse, but neither constitutes independent semantic ground truth.

[1] §3.2 and §4.1.6 · mean glossary-classifier accuracy 94.5%; C4 is 70.8%

One filtered precondition subset trains one targeted LoRA.

FROZEN BACKBONE

W

FLUX.1-dev: a pretrained 12-billion-parameter image generator.

+

TRAINABLE UPDATE

BA

Small low-rank matrices adapt the model to one filtered subset.

$$W' = W + BA$$

The original weights W stay frozen; inference uses the adapted weights W′ after adding the learned low-rank update BA.

BENEFIT	Generate many new candidates inside a rare requirement region
EVIDENCE SOUGHT	Relevance, distributional plausibility, and semantic variation
NOT ESTABLISHED	Non-memorization or systematic coverage within that region

[1] §3.3 and §4.1.4 · LoRA equation simplified; latent-space coverage remains future work

The six research questions validate inputs before interpreting model outcomes.

A credible failure is a joint event: the generated input satisfies the precondition, but the model output violates the postcondition. RQ = Research Question.

Validate the generated inputs

RQ1–RQ3

RQ1	Consistency	Does the image satisfy the precondition?
RQ2	Plausibility	Is its distribution close to matching training images?
RQ3	Variation	Are other glossary features distributed similarly?

Evaluate the resulting tests

RQ4–RQ6

RQ4	Comparison	Are targeted outcomes more informative than baselines?
RQ5	Violations	Do tests reveal credible postcondition failures?
RQ6	Diagnosis	Do failures expose repeatable decision patterns?

[1] §4.1 · input validity is established before model outcomes are interpreted

Targeted LoRAs satisfy the requested precondition 92.1% of the time on average.

92.1%

mean across 21 requirements in MNIST, CelebA-HQ, and SGSM.

M3 · targeted LoRA

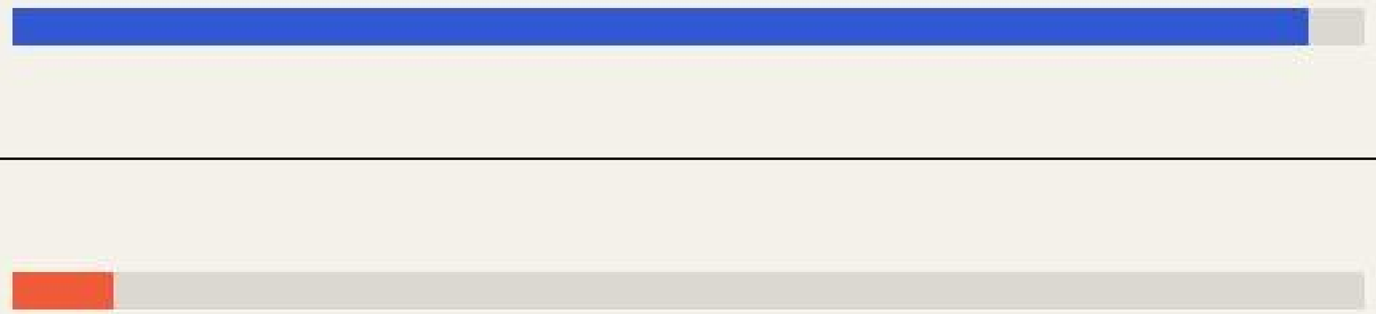
filtered images: very thick 7

95.8%

M3 · one all-data LoRA

all annotated MNIST training images; not plain FLUX

7.4%



COMPARISON SHOWN

Precondition-specific LoRA versus one LoRA trained on all annotated MNIST data · 100 images × 10 repetitions.

BOUNDARY

Glossary classifiers average 94.5% accuracy; they are measurement proxies, not semantic ground truth.

KID compares each generator with real images satisfying the same precondition.

MNIST · M3 · KID ↓

.308 → .039

PROMPT-ONLY FLUX

.308

TARGETED LORA

.039

SGSM · S2 · KID ↓

.242 → .038

PROMPT-ONLY FLUX

.242

TARGETED LORA

.038

CELEBA-HQ · C1 · KID ↓

.123 → .097

PROMPT-ONLY FLUX

.123

TARGETED LORA

.097

KID · KERNEL INCEPTION DISTANCE

A distance between two image sets in a learned feature space · lower means the distributions are closer.

BOUNDARY

KID is not an image-level realism probability; several reference subsets contain fewer than 300 images.

[1] §4.2.2 and Table 3 · examples show KID(reference, prompt-only FLUX) → KID(reference, targeted LoRA)

Against the same filtered training subset, targeted LoRA preserves the feature mix better than prompt-only FLUX.

MNIST · MEAN JS	CELEBA-HQ · MEAN JS	SGSM · MEAN JS
7.2× closer	1.4× closer	5.5× closer
PROMPT-ONLY FLUX .1765	PROMPT-ONLY FLUX .1150	PROMPT-ONLY FLUX .1960
TARGETED LORA .0244	TARGETED LORA .0831	TARGETED LORA .0355

WHAT "FEATURE FREQUENCY" MEANS

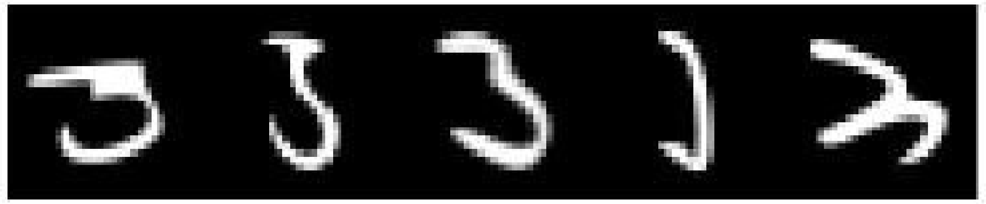
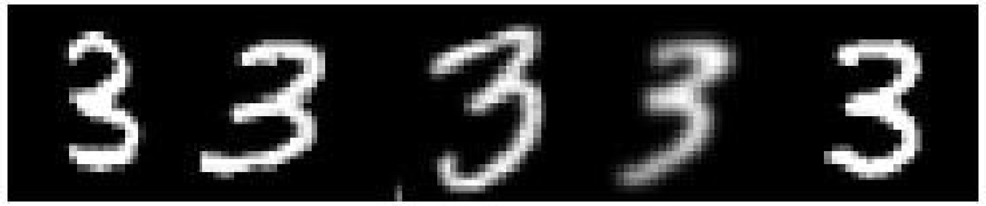
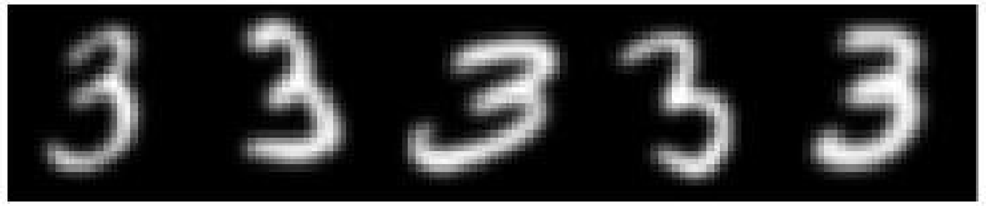
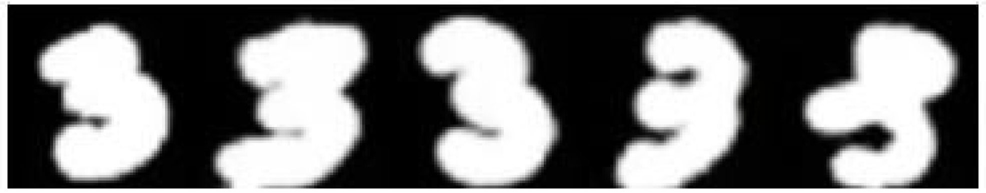
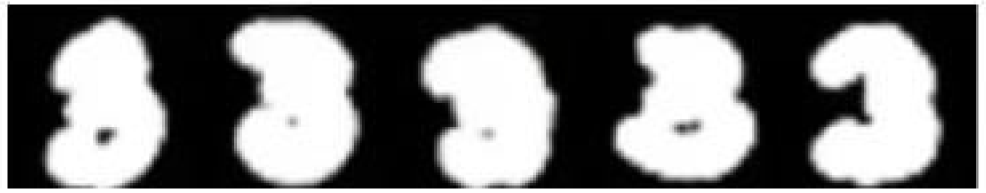
Example: among thick 7s, how often are images also left-leaning, right-leaning, tall, or short? Compare those label proportions with the filtered training subset.

JS · JENSEN-SHANNON DIVERGENCE

One distance between those two label distributions · lower means closer. It does not measure pixel diversity, memorization, or latent-space coverage.

[1] §4.2.3 and Table 4 · displayed domain means are calculated from the seven reported rows

M2 requirement: if the image is a very thick 3, the classifier must predict 3.

Technique	Passed Tests	Failed Tests
Deephyperion		
Image Transform		
RBT4DNN		

Original paper evidence · Table 5 · cropped only for legibility · CC BY 4.0.

WHAT PASS / FAIL MEANS HERE

For M2, pass = the MNIST classifier predicts “3”; fail = it predicts another digit. The rows separate those outputs for each generator.

INTERPRETATION RULE

DeepHyperion and transformations mostly miss “very thick”; therefore neither their passes nor failures are evidence about M2.

RBT4DNN exposed failures broadly, and the audited subset was mostly valid.

AUTOMATED OUTPUT CHECK

19/21

requirements

The model under test violated the encoded postcondition at least once.

→

HUMAN INPUT AUDIT

11×30

sampled failures

Multiple authors checked realism and precondition validity.

→

AUDITED QUALITY

3.3%

mean false positives

330 images total; 11.5% worst case.

WHO DECIDES WHAT?

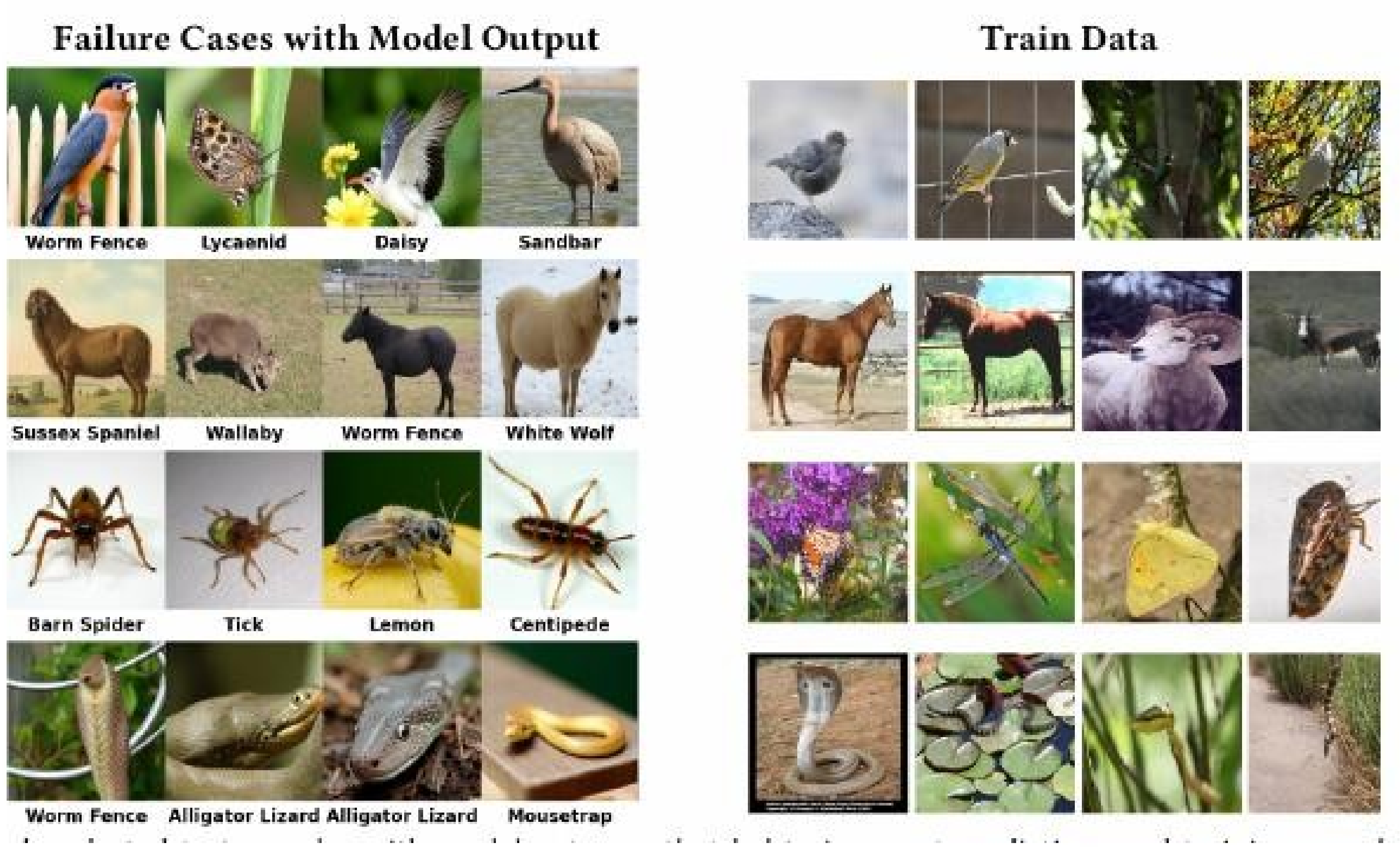
Code checks the model output against the postcondition; humans inspect 330 selected failing images for realism and precondition match.

WHERE VQA/GTCS ENTER

Automated semantic labels support the broader pipeline, but the reported RQ5 false-positive audit is multi-author human assessment.

[1] §4.1.6 and §4.3.2 · assessor disagreement on precondition consistency was 1.7%

Across four animal requirements and three classifiers, failures reveal recurring model-specific patterns.



Original paper Table 7 · generated failure cases and training examples · cropped for legibility · CC BY 4.0.

METHOD

4 requirements × 3 classifiers × repeated generation

Collect 1,486 postcondition failures, rank frequent wrong labels, then manually inspect associated images.

EXAMPLE

A bird near a fence → “worm fence”

Other cases align with insects, horses, or snakes that resemble frequent training classes or backgrounds.

BOUNDARY

Association, not causal diagnosis

The study does not change only the background while holding the animal fixed.

The paper demonstrates feasibility—not a complete assurance process.

GENERATIVE TRAINING

Better tests

Improve generator training and adaptation to reveal more observed failures and reduce false positives.

COVERAGE EVIDENCE

Latent-space coverage

Map the model's internal feature space and deliberately search regions that generated tests have not exercised.

GENERALIZATION

More domains

Expand driving-like safety datasets and explore broader robustness and functional requirements.

ASSURANCE BOUNDARY

The comparator changes with the question; automated labels, narrow baselines, and small KID subsets limit how far the evidence generalizes.

[1] §4.4 and §5 · author-stated scope kept separate from this seminar's critique

A model error counts as requirement evidence only if the generated image really satisfies the requirement.

PRECONDITION INVALID · MODEL PASSES	PRECONDITION INVALID · MODEL FAILS
Irrelevant pass The output says nothing about the stated requirement.	**Off-target false positive** A model error is not yet a requirement counterexample.
PRECONDITION VALID · MODEL PASSES **Valid requirement pass** Relevant evidence, but never proof of the universal property.	PRECONDITION VALID · MODEL FAILS **Credible counterexample** This joint event is the paper's strongest testing evidence.

SAMPLED JOINT AUDIT

30 failures × 11 low-pass requirements = 330 independently assessed samples.

Critique in one sentence: the authors manually verify 330 selected failures, but most reported results still depend on imperfect automated labels—so confidence is strongest for the audited sample, not the full failure population.

Sharing one MNIST LoRA saves training—but sharply weakens targeting on the hardest requirements.

METHOD

Reuse the authors' MNIST ablations

Compare precondition-match rates for a targeted LoRA, one LoRA over six requirement subsets, and one LoRA over all MNIST data.

6 requirements

EASY TO SHARE · M4

Targeted 92.5% · shared 56.4%

The shared requirement LoRA remains usable, though substantially less precise. The all-data LoRA reaches only 12.7%.

training ↘ · targeting ↘

HARD TO SHARE · M3

Targeted 95.8% · shared 27.5%

The all-data LoRA reaches 7.4%. A single generator does not preserve difficult semantic regions automatically.

targeting matters

EXTENSION MESSAGE

Per-requirement LoRAs are not merely an implementation detail: they buy semantic precision. The open design question is whether conditioning can recover that precision in one shared model.

One conditional VAE matched digit-pass rates—but did not test requirement validity.

VAE = Variational Autoencoder: an encoder compresses images into a probabilistic latent space; a decoder samples that space to generate images.

TRAINING DATA

Copy upstream RBT4DNN outputs

Use per-requirement generated MNIST images and attach the requirement ID M1–M6.

→

ONE NEW MODEL

Train one conditional VAE

The requirement ID conditions generation. This model is neither FLUX nor LoRA and does not use original MNIST directly.

shared generator

→

EVALUATION

Generate and classify digits

Three seeds × 100 images per requirement; evaluate only the digit postcondition with a classifier.

mean .945 · M3 .743

ONLY THE COMMON METRIC CAN BE COMPARED

Mean digit pass: per-requirement LoRAs .942 ↔ conditional VAE .945 · M3: .724 ↔ .743. VAE precondition match was not measured, so this is a feasibility signal—not evidence that it replaces LoRA.

≈24.7 serial A100-hours motivate a controlled shared-generator study

[1] Fig. 3 · seminar mnist-shared-generator · no direct precondition, KID, JS, or human-validity evaluation

Requirements make tests relevant. Validity makes failures credible.

WHAT THE PAPER ESTABLISHES	WHAT LIMITS THE CLAIM
Structured requirements can guide distributionally plausible, glossary-diverse, scenario-specific test generation across four domains.	Semantic labels and glossary classifiers are useful proxies; credible failures still depend on joint input-and-output validation.

References.

[1] **N. J. Mozumder, F. Toledo, S. Dola, and M. B. Dwyer.** "RBT4DNN: Requirements-based Testing of Neural Networks." arXiv:2504.02737, v3, 2025. CC BY 4.0.

[2] **E. J. Hu et al.** "LoRA: Low-Rank Adaptation of Large Language Models." ICLR, 2022.

[3] **RBT4DNN artifact.** github.com/less-lab-uva/RBT4DNN · seminar provenance pinned to commit c3deff871ed41043d83a05289faab84c41706586.

[4] **Seminar experiments.** Reproducible notebooks, results, summaries, and provenance in this repository.

Original paper ↗	Original artifact ↗